Lactase Persistence by Population

Background

Lactase persistence (LP) involves continued expression of lactase-phlorizin hydrolase (LPH) through the individuals' lifetime, maintaining high LPH levels (Leseva et al., 2018). Lactase persistence and non-persistence are associated with a T/C polymorphism at position −13910 upstream the lactase gene as well as polymorphisms at -22018 G>A (Lewinsky et al., 2005, Thammachai et al., 2024). The variations can be attributed to positive selection in populations with a history of dairy farming, including Northern Europeans, West African groups such as the Fulani in the Savannah region, and North Africans populations in the Maghreb region (Vicente et al., 2019, Campbell et al., 2021).

The LPH enzyme is encoded by the lactase (LCT) gene, located on the chromosome 2q21, is comprised of 17 exons and covers about 49kb. Based on previous studies, it has been theorized that LP could be independently caused by at least five single nucleotide polymorphisms (SNPs) that sit ~14kb and ~22kb upstream of LCT in a regulatory region called *MCM6* (minichromosome maintenance complex component 6), which is located upstream with a transcriptional start site that is 39 kb 5' from the LCT gene and ending 3.5 kb from the start site the LCT gene (Enattah et al., 2002, Anguita-Ruiz et al., 2020). While these two genes are close in the genome, current evidence shows that their regulation is likely independent of each other (Harvey et al., 1996). LP in Eurasia is tightly associated with derived alleles T at rs4988235 and A at rs182549 SNPs that are located -13,910 bp and -22,018 bp, respectively, upstream of the LCT gene (Satta et al., 2020). Variations in CC genotype at rs4988235 and the GG at rs182549, are associated with low lactase activity, which is predominant in East Asian populations, particularly Japan and China (Kato et al., 2018).

Dataset Source and 1000 Genomes Project

Data for this study was obtained from the 1000 Genomes Project which was designed to identify common genetic variants occurring at a frequency of at least 1% in the population (IGSR, 2014). The dataset used in this project was obtained from The Phase 3 which introduced an expanded set of variant callers 2,504 samples mapped on GRCh37 in 2014, which was later expanded to include structural variants mapped to GRCh38, increasing the sample size to 3,202. The coordinates of the selected Lactose Persistence SNPs span chromosome 2 positions 135,700,000 to 136,800,000, covering introns 13 and 9 that contain rs4988235 at position 136,608,646 and rs182549 at position 136,616,754. Before filtering, this region included 26,319 variants.

Three superpopulations were included in this analysis: African (AFR = 893), East Asian, (EAS = 585), and European (EUR = 633), totaling to 2111 individuals. South Asians and Admixed Americans were excluded to focus on the three groups with the most distinct LP frequency and to reduce population heterogeneity. SNP quality control included minor allele frequency filtering

to retain SNPs with MAF > 0.001 to remove rare reference and rare alternate alleles, leaving 12,877 informative SNPs. Hardy Weinburg equilibrium was initially performed per population but was ultimately excluded from analysis as it removed biologically important variants (Pearman et al., 2022; Hsu et al., 2026). Variant density was broadly uniform across the 1.1Mb window with no coverage gaps.

Objectives

The goal of this study is to evaluate and compare the effectiveness of unsupervised clustering models (PCA and K-means) versus supervised machine learning model (Random Forest) in classifying individuals from AFR, EAS, and EUR superpopulations using single-nucleotide polymorphism genotype data from the loci of *LCT* and *MCM6*.

Methods

Principal component analysis (PCA) was applied to genotype matrix with mean-centering and unit-variance scaling. Scaling ensures that rare variants can contribute evenly to the principal components to avoid them being dominated by high variance SNPs. The top 11 PCs were retained for clustering, selected based on the inflection point of the scree plot where the rate of variance explained per additional PC substantially decreased.

K-means clustering (*nstart=25*) was applied to the top 11 PC scores. The optimal number of clusters was determined using the elbow and silhouette methods; the elbow plot showed no clear inflection point, while silhouette analysis indicated *k=6* as the best balance between cluster cohesion and separation. Cluster assignments were tabulated against true superpopulation labels to evaluate separation quality.

A random forest (RF) classifier was trained on an 80/20 stratified training and testing split (n=1690 training n=421 testing). The *mtry* hyperparameter was tuned with *mtry=226* minimizing out of bag error to 1.89%. The final model used *ntree = 500*. Performance was evaluated using overall accuracy, Cohen's Kappa, and per-class F1-score. Variable importance was assessed via mean decrease in accuracy (MDA) and mean decrease in Gini impurity (MDG). A pairwise linkage disequilibrium (LD) back-check was performed on the top RF and PCA SNPs to determine if these SNPs were representing independent sources of information, or if they were tagging the same underlying haplotype.

Results

Initial data exploration **(Figure 1)** showed that there was some class imbalance between superpopulations of interest. Specifically, AFR has many more samples than EUR and EAS. The variation distribution showed no major coverage gaps. The heterozygosity plot showed that there were some outliers in heterozygosity in the EAS superpopulation. These are likely rare mutation events that were removed with the minor allele frequency filtering.

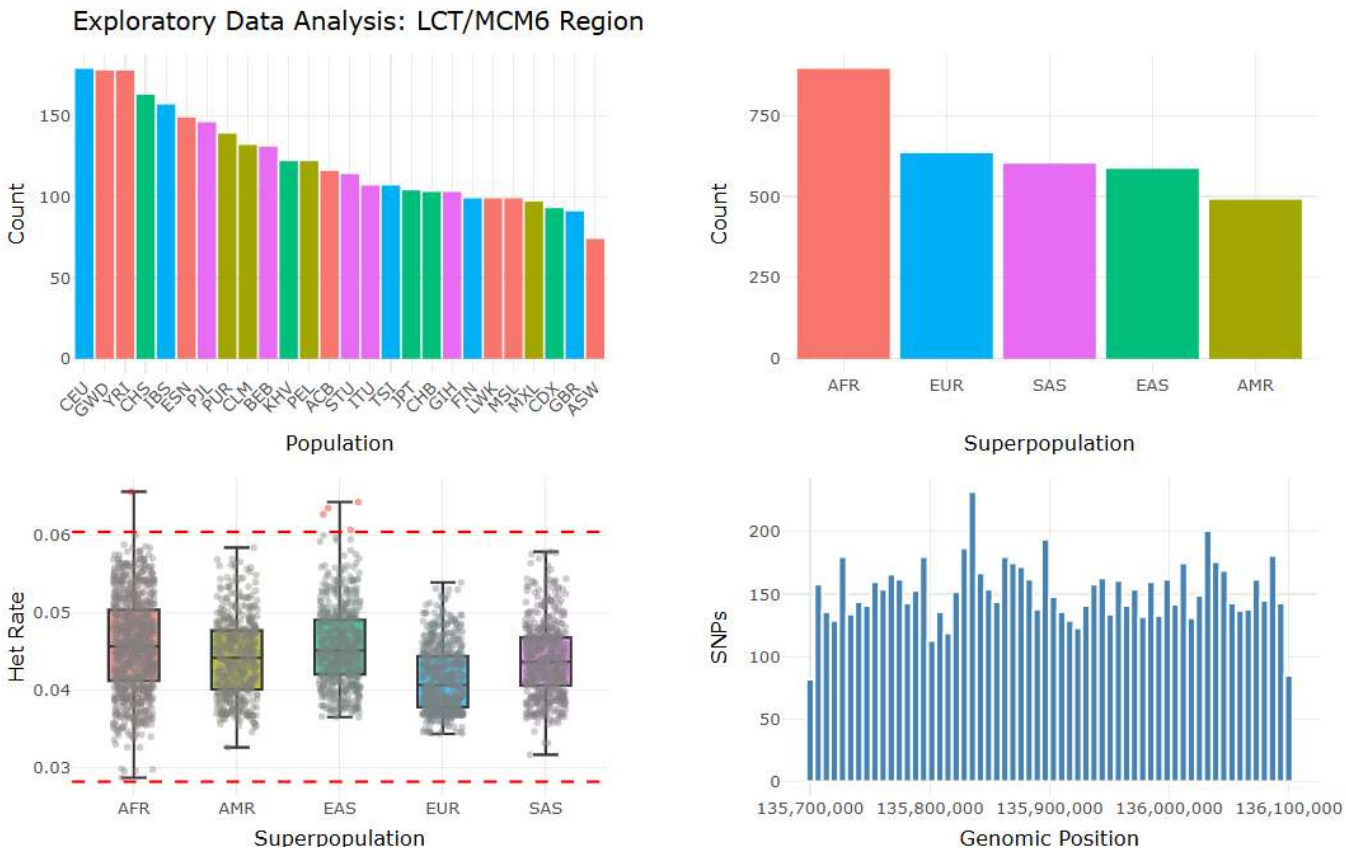

Figure 1. Exploratory Data Analysis: Top Left: Bar plot of subpopulations. Top Right: Bar plot of superpopulations. Bottom Left: Heterozygosity Plot. Bottom Right: Variation Distribution across chromosome.

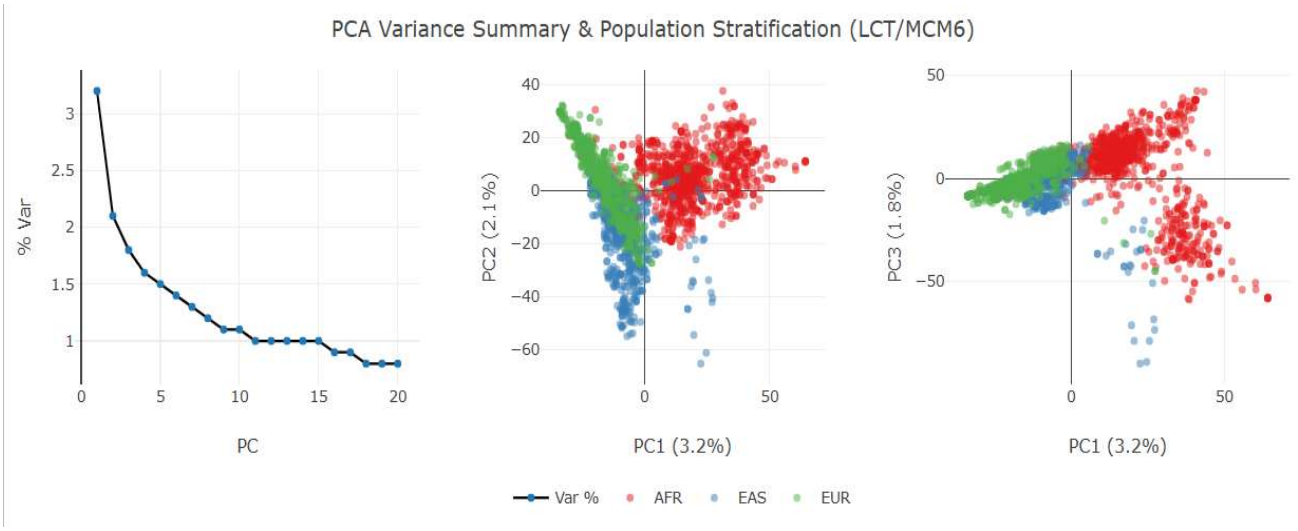

Figure 2. PCA variance summary and population clustering based on *LCT* and *MCM6* loci.

Figure 2A is a Scree plot illustrating the distribution of genetic variance across the principal components (PC). PC1 accounted for 3.2% of the total variance, followed by PC2 (2.1%) and PC3 (1.79%). The Scree plot inflection point indicates that the most variation is captured by the first 11 principal components, which was used to inform the number of PCs to include in the K-means clustering. This ensures that the unsupervised models are trained using informative genetic variation. PC1, PC2 and PC3 (**Figure 2B & 2C**) mostly clustered the three superpopulations (AFR, EAS, EUR), though there is significant overlap between EUR and EAS clusters. PCA result identified: *2:136431332:A:G* and *2:136431325:A:ATT* as the highest loading SNPs for PC1.

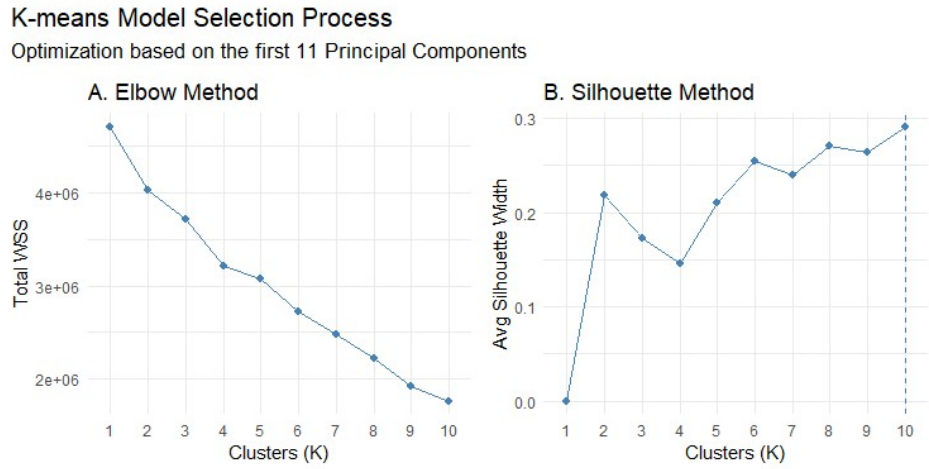

Figure 3: K-means model selection for optimal number of clusters based on LCT and MCM6 loci

The elbow-plot (**Figure 3A**) showed no bend, which is likely due to the high-dimensional genomic data. The Silhouette method (**Figure 3B**) maps both internal cluster distance and separation from nearest neighboring clusters. It found that the optimal K was 10, but showed a similar peak around 6 clusters, which was chosen for K-means clustering to avoid overfitting.

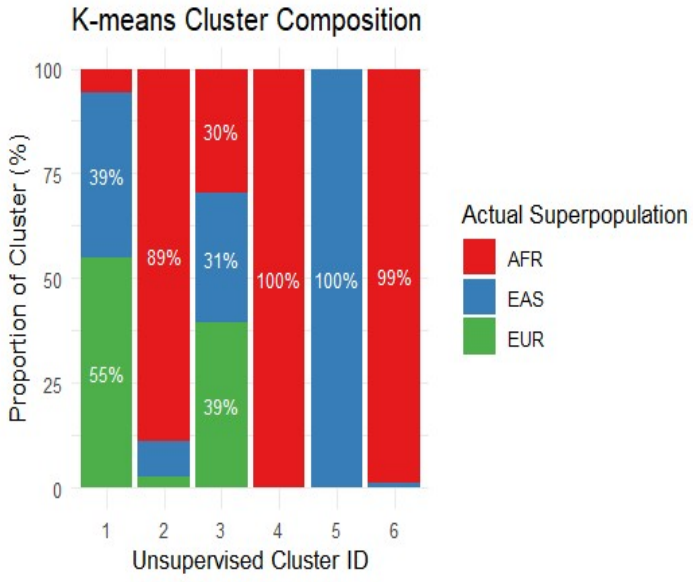

Figure 4: K-means clustering composition of AFR, EAS, and EUR superpopulations based on LCT and MCM6 loci

The K-means cluster compositions (**Figure 4**) showed that clusters 2, 4, 5, 6 were very well clustered for superpopulations; cluster 1 and 3 are catch-all groups containing samples from each superpopulation. This suggests that their specific haplotypes at the LCT and MCM6 loci are divergent, while others remain ancestral.

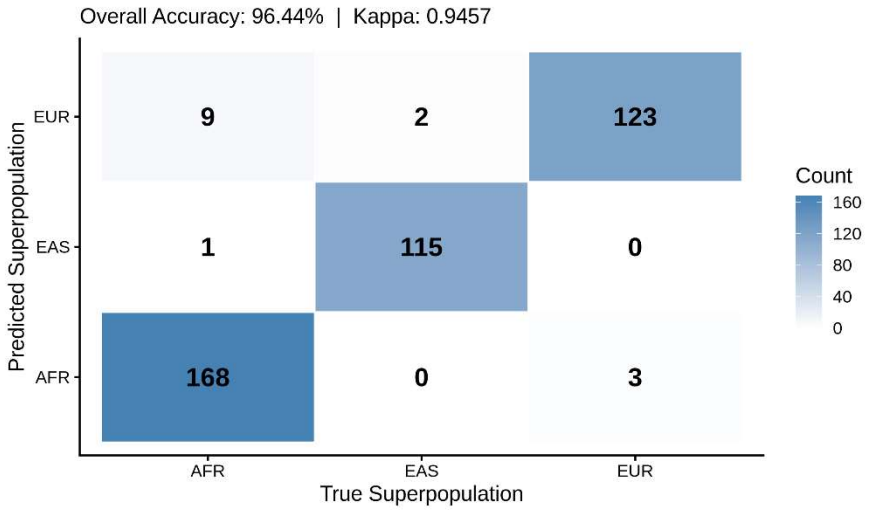

Figure 5. Confusion matrix of random forest classification for AFR, EAS, and EUR superpopulations based on LCT and MCM6 SNP genotypes.

The Random Forest classifier achieved 96.44% overall accuracy and a Cohen's Kappa of 0.9457 on the held-out test set (**Figure 5**), indicating strong agreement between predicted and true superpopulation labels. Per-class F1 scores were 0.9871 for EAS, 0.9628 for AFR, and 0.9462 for EUR, with AFR and EAS showing the highest classification accuracy and EUR exhibiting slightly more misclassification due to overlap with the other groups (precision = 0.9179). Variable importance analysis identified *2:136431325:A:ATT* and *2:136431332:A:G* as the top two SNPs by both MDA (8.26 and 7.32) and MDG (20.51 and 16.62). Pairwise LD analysis confirmed that these two SNPs are in near-perfect linkage with each other (r^2=0.998), while a mean r^2 of 0.171 across all 28 analyzed SNPs suggests additional independent signals distributed across the wider region.

Comparison of Methods

K-means partially resolved population structure at the *LCT/MCM6 loci*, with several clusters strongly enriched for individual superpopulations, but AFR individuals were dispersed across multiple clusters rather than forming a single coherent group. RF demonstrated that finer population distinctions can be learned from this locus when labels are available, achieving 96.44% accuracy. Additionally, RF achieved near-perfect accuracy in classifying EAS individuals (F1=0.987), which K-means could not replicate. Notably, both methods independently identified the same two *MCM6* variants as the dominant signals of population differentiation, strengthening confidence in the biological relevance of this region. These findings suggest that unsupervised approaches can detect broad ancestry structure, but supervised classification is required for more precise individual-level assignment.

Discussion

Genetic variation in the LCT/MCM6 region captures clear population structure across the 2,111 individuals analyzed. As shown in **Figure 2**, PCA separates AFR individuals distinctly from EUR and EAS along PC1, while EUR and EAS remain partially overlapping across PC2 and PC3. This reflects known differences in lactase persistence allele frequencies across populations

and indicates that genetic variation at this locus is distributed across multiple dimensions rather than dominated by a single axis (Anguita-Ruiz et al., 2020; Enattah et al., 2002).

Clustering results further support this pattern. The K-means model selection process in **Figure 3** shows no clear elbow point and a relatively flat silhouette profile, highlighting the difficulty of defining discrete clusters in high-dimensional genomic data. This ambiguity is reflected in the final clustering output, where **Figure 4** demonstrates that while several clusters are strongly enriched for individual superpopulations, others act as mixed groups containing AFR, EAS, and EUR individuals. This suggests that only a subset of haplotypes at the LCT/MCM6 locus are strongly distinguishable, while others remain shared across populations (Vicente et al., 2019).

In contrast, supervised learning was able to resolve these finer distinctions. As shown in **Figure 5**, the RF classifier achieved high predictive performance (96.44% accuracy; $\kappa = 0.9457$), with near-perfect classification of AFR and EAS individuals and only minor misclassification in EUR. This demonstrates that although population structure is not fully separable in low-dimensional PCA space or through clustering alone, it can be recovered when models integrate signals across multiple SNPs.

Both unsupervised and supervised approaches converged on the same *MCM6* regulatory region as the primary driver of differentiation. The top SNPs identified by PCA loadings and RF importance fall within this region, which contain enhancer elements regulating *LCT* expression (Lewinsky et al., 2005; Harvey et al., 1996; Leseva et al., 2018). Linkage disequilibrium patterns further support a structured haplotype shaped by positive selection (Satta & Takahata, 2020). Overall, the LCT/MCM6 locus is highly informative for broad ancestry differentiation, particularly separating AFR from Eurasian populations, but finer distinctions, especially between EUR and EAS, require supervised models reflecting the complex and population-specific evolution of lactase persistence (Thammachai et al., 2024; Anguita-Ruiz et al., 2020).

A key limitation of this study is the focus on a single genomic region, which restricts the ability to capture genome-wide population structure and may exaggerate the influence of locus-specific selection. The presence of heterozygosity outliers in EAS populations may indicate a high mutation rate in a population or a heterozygote advantage, which would be ineffective at representing overall lactase persistence in a superpopulation. Additionally, there is some class imbalance between superpopulations, with AFR contributing a larger number of samples than EUR and EAS (**Figure 1**). While not extreme, this imbalance may influence model performance, particularly in supervised learning, by biasing classification toward the majority class. The exclusion of admixed (AMR) and South Asian (SAS) populations further simplifies classification but limits the generalizability of the findings to more complex, real-world population structures. While RF achieved high accuracy, its performance may partly reflect strong linkage disequilibrium within the region rather than fully independent predictive signals. Future work could extend this analysis to genome-wide SNP data, incorporate additional

populations, and apply methods such as admixture modeling or haplotype-based approaches to better capture fine-scale structure and evolutionary history.

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Contributions Table and Task List

Tasks:

1. Data preprocessing & EDA: Define slice of chromosome and get vcf, get minor allele frequencies, superpopulations of interest, filter data.
2. PCA implementation & visualization: Conduct PCA, plot, visualize
3. Interpretation of PCA results: Results and figures
4. Random Forest model development: Develop 80/20 train-test split, tune for hyperparameters, update model, track out-of-bag error.
5. Model evaluation and performance
6. Feature importance & SNP interpretation: Contrast MDA and MDG, find informative SNPs from both lists
7. Literature review & introduction writing: Get background on genetics of lactase persistence and superpopulations affected
8. Methods writing: illustrate the methods used to conduct analyses and parameters
9. Results writing & figures: Plot figures, explain results, labels and captions
10. Discussion & conclusions: Consolidate results with background on genetics of lactase persistence
11. Code consolidation & script preparation: Merge scripts and models into one .R file.
12. Final editing & formatting: Organize report and tasks list.

Task	Maryanne (1)	Rameen (2)	Stephanie (3)	Matei (4)
1	C(%) = 10 Assisted with dataset checks and summary interpretation	C(%) = 10 Assisted with data review and EDA validation	C(%) = 70 Did data download, inspection, filtering decisions, and biological context	C(%) = 10 Reviewed preprocessing steps and outputs
2	C(%) = 10 Reviewed PCA outputs and interpretation	C(%) = 10 Assisted with implementation troubleshooting	C(%) = 70 PCA implementation, visualization, and variance analysis	C(%) = 10 Led Assisted with visualization formatting
3	C(%) = 10 Provided biological context support	C(%) = 10 Assisted with interpretation discussion	C(%) = 70 Led interpretation and integration with biological context	C(%) = 10 Assisted with result validation

4	C(%) = 10 Reviewed modeling approach	C(%) = 70 Implemented model and parameter tuning	C(%) = 10 Assisted with model setup and validation	C(%) = 10 Assisted with model interpretation and output review
5	C(%) = 15 Reviewed evaluation metrics	C(%) = 35 Calculated performance metrics	C(%) = 35 Assisted with validation and result checking	C(%) = 15 Interpreted performance outputs
6	C(%) = 10 Provided biological insight on key variants	C(%) = 70 Led feature importance analysis	C(%) = 10 Assisted with result validation	C(%) = 10 Reviewed interpretation
7	C(%) = 70 Conducted literature review, wrote introductory segment	C(%) = 10 Assisted with finding papers for literature review	C(%) = 10 Helped with understanding and with background wording	C(%) = 10 Reviewed, checked grammar, suggested improvements
8	C(%) = 5 Reviewed methods for biological accuracy	C(%) = 70 Wrote Methods section	C(%) = 15 Edited for correctness, clarity and formatting	C(%) = 10 Reviewed and edited content
9	C(%) = 5 Reviewed results for consistency	C(%) = 20 Wrote initial comparison paragraphs, wrote RF, MDA and MDG sections	C(%) = 70 Wrote Results section and prepared figures	C(%) = 5 Reviewed figures and narrative flow
10	C(%) = 10 Provided biological context input	C(%) = 10 Assisted with interpretation discussion	C(%) = 10 Reviewed conclusions for clarity	C(%) = 70 Wrote synthesis, discussion, and conclusions
11	C(%) = 15 Reviewed code structure	C(%) = 35 Changed kmeans to use top 11 PCs, compared with HC	C(%) = 35 Wrote initial code, dataset downloading script, PCA creation, wrote Kmeans, finalized script	C(%) = 15 Reviewed and finalized script

12	C(%) = 25 Collected references, created bibliography	C(%) = 10 Assisted with final checks	C(%) = 10 Reviewed formatting and figures	C(%) = 45 Final editing and formatting, down to 8 pages